Qatar-1b

R-1857

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_141215` · created 2026-08-03T08:55:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457006.59654 +/- 0.00044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.052
Transit depth (Rp/Rs)^2	3.68 +/- 2.0 [%]
Orbital Inclination (inc)	82.63 +/- 2.38
Airmass coefficient 1 (a1)	1.01 +/- 0.017
Airmass coefficient 2 (a2)	-0.002 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.74 %
Best Comparison Star	#2 - [320, 90]
Optimal Method	PSF photometry
Transit Duration (day)	0.045 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.052 vs 0.1463 (deviation 0.88σ)
Duration fit vs published	0.045 d vs 0.0692 d (deviation 1.73σ)
Mid-time O–C	0.2 min
Depth SNR	3.7
Residual scatter	1.74 %

Light curve

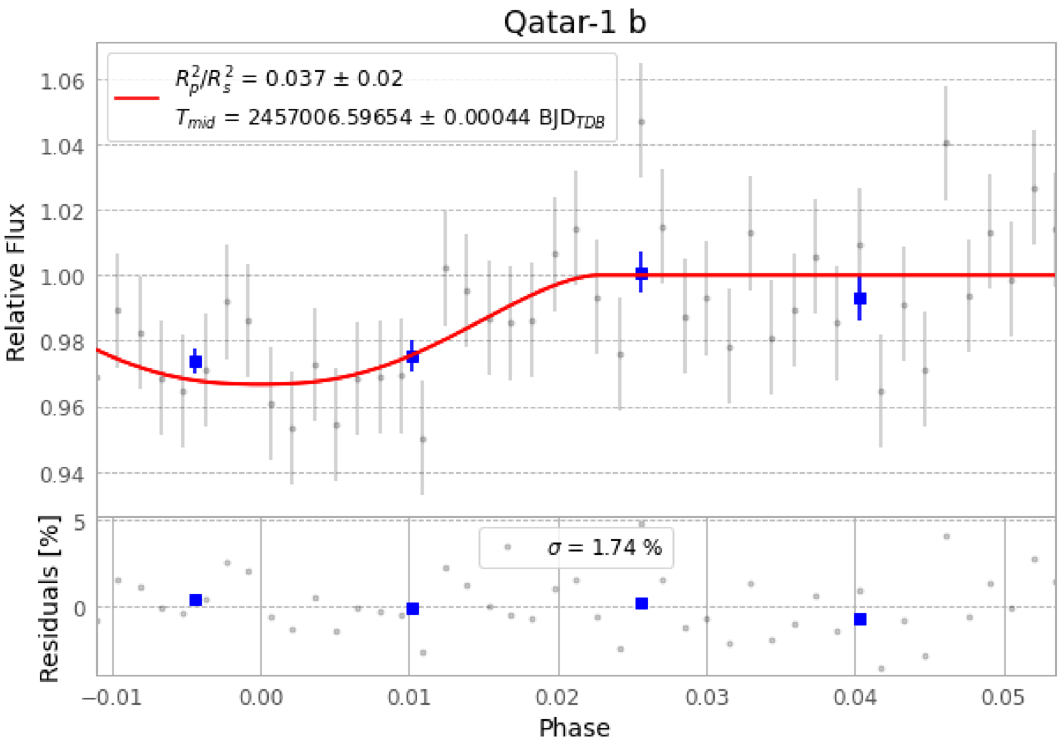

Figure 1 — FinalLightCurve_Qatar-1 b_2014-12-14.png (SHA-256 6e22ae6c0d0fa53c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 244], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 785bf10327b1cce6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

45 FITS frames (30 MB), `Qatar-1141215015412.FITS` → `Qatar-1141215040613.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-141215.log (fda1953025ce…), FinalLightCurve_Qatar-1 b_2014-12-14.png (6e22ae6c0d0f…), FinalLightCurve_Qatar-1 b_2014-12-14.csv (00f7b952bbd4…)

Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)